

The Complete Spectrum of the Fair Lamplighter Chain on a Finite Cycle

Source-local theorem package HCS-C341

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Abstract

We diagonalize the fair switch–walk–switch chain on binary lamps over every finite cycle. Walsh transformation turns switching into deletion projections; each nontrivial block is a direct sum of killed lazy paths. This yields the full characteristic polynomial, eigenbasis, multiplicities, sharp spectral gap and exact boundary cases. Finite enumeration is used only as a regression receipt. No arithmetic target or Route-B claim is made.

1 Frozen model

Write $C_n = \mathbb{Z}/n\mathbb{Z}$ and $\Omega_n = \mathbb{Z}_2^{C_n} \times C_n$. For $n \geq 3$ the base kernel Q_n stays with probability $1/2$ and moves to each neighbour with probability $1/4$. At $n = 2$ the two directed choices coalesce, so the other vertex has probability $1/2$; $Q_1 = (1)$. One chain step independently replaces the lamp at the departure site by a fair bit, applies Q_n , and independently replaces the arrival lamp by a fair bit. Replacement, not deterministic toggling, is part of the definition.

For $A \subseteq C_n$ define the lamp character $\chi_A(\eta) = (-1)^{\sum_{x \in A} \eta(x)}$ and the position projection $D_A(x, x) = \mathbf{1}_{\{x \notin A\}}$. If R_A denotes the multiset of positive lengths of the cyclic runs in $C_n \setminus A$, put

$$c_{n,k} = \frac{1 + \cos(2\pi k/n)}{2}, \quad p_{\ell,j} = \frac{1}{2} + \frac{1}{2} \cos \frac{\pi j}{\ell + 1}.$$

Theorem 1 (Full finite-cycle spectrum). *The uniform law on Ω_n is the unique reversible stationary law and*

$$P_n \cong \bigoplus_{A \subseteq C_n} D_A Q_n D_A. \quad (1)$$

The empty-support block has eigenvalues $c_{n,k}$, $0 \leq k < n$. Every $A \neq \emptyset$ block has $|A|$ zero eigenvalues and, for each $\ell \in R_A$, the simple eigenvalues $p_{\ell,j}$, $1 \leq j \leq \ell$. Walsh characters times deleted-site deltas and path sines form a complete orthogonal eigenbasis. Consequently

$$\det(zI - P_n) = \det(zI - Q_n) \prod_{\emptyset \neq A \subseteq C_n} z^{|A|} \prod_{\ell \in R_A} \prod_{j=1}^{\ell} (z - p_{\ell,j}). \quad (2)$$

For $n \geq 3$ the largest nonconstant eigenvalue has multiplicity n and

$$\lambda_{\text{top}} = \frac{1 + \cos(\pi/n)}{2}, \quad \text{gap}(P_n) = \frac{1 - \cos(\pi/n)}{2}. \quad (3)$$

For every mean-zero f , $\|P_n^t f\|_2 \leq \lambda_{\text{top}}^t \|f\|_2$, and equality is attained. At $n = 2$ the gap is $1/2$; at $n = 1$ the spectrum is $\{1, 0\}$ and the gap is one.

2 Walsh blocks

Let S be one fair switch. It is conditional expectation over the current lamp, hence an orthogonal projection in uniform L^2 . The symmetric kernel Q_n is self-adjoint, so $P_n = S Q_n S$ is self-adjoint and preserves constants. A positive-probability path can visit every site and choose prescribed switch outcomes; the finite chain is irreducible, proving uniqueness.

On $\chi_A(\eta)g(x)$, fair averaging gives

$$S(\chi_A g) = \chi_A D_A g :$$

the character is unchanged when $x \notin A$ and has mean zero when $x \in A$. The position move leaves the lamp character fixed. Thus switch, move and switch acts as $D_A Q_n D_A$, proving (1).

3 Path diagonalization and completeness

For $A = \emptyset$, cyclic Fourier vectors give $c_{n,k}$. If $A \neq \emptyset$, the coordinates in A are killed and supply $|A|$ zero vectors. Every remaining run of length ℓ carries the tridiagonal matrix with diagonal $1/2$ and neighbouring entries $1/4$. With Dirichlet endpoints,

$$u_j(r) = \sin \frac{\pi jr}{\ell + 1}, \quad 1 \leq r \leq \ell,$$

satisfies

$$\frac{1}{4}u_j(r-1) + \frac{1}{2}u_j(r) + \frac{1}{4}u_j(r+1) = p_{\ell,j}u_j(r).$$

Discrete sine orthogonality gives ℓ vectors on each run. Together with the deleted deltas there are $|A| + \sum_{\ell \in R_A} \ell = n$ vectors in sector A ; distinct sectors are Walsh-orthogonal. The basis and product formula (2) follow.

4 The sharp mode and small cycles

All eigenvalues are nonnegative. The empty sector's largest value below one is $(1 + \cos(2\pi/n))/2$. A nonempty support leaves runs of length at most $n-1$, while $(1 + \cos(\pi/(\ell+1)))/2$ increases strictly with ℓ . The maximum below one is therefore (3), attained exactly by singleton supports. Each corresponding path top eigenvalue is simple, so its multiplicity is exactly n . The spectral theorem gives the stated contraction and an eigenfunction gives equality.

For $n=2$, the empty block is $\{1,0\}$, each singleton-support block is $\{1/2,0\}$, and the full-support block is zero. For $n=1$, the position is fixed and the fair lamp replacement has spectrum $\{1,0\}$. These computations also show why directed-neighbour multiplicities must be coalesced before forming Q_2 .

Round-one proof strengthening. The previous paragraphs add the complete eigenbasis, the uniqueness of the top-mode owner, its multiplicity, sharpness, and both degenerate-cycle cases.

5 Source lineage

The general switch-walk-switch/percolation correspondence is due to Lehner, Neuhauser and Woess, *Math. Ann.* 342 (2008), 69–89, doi:10.1007/s00208-008-0222-7. Graph-level eigenspaces were developed by Lehner, *Proc. AMS* 137 (2009), 2631–2637, doi:10.1090/S0002-9939-09-09869-4. The present finite-cycle theorem is a self-contained specialization and makes no priority claim for that general theory.